

COMICCRAFT

DOCUMENT 19: SAMPLE PROJECT DOCUMENTATION (ComicCraft Full Reference)

Project Description

ComicCraft is an AI-powered web application that leverages Google Gemini 1.5 Flash & Pro models alongside Stable Diffusion / FLUX.1-schnell for intelligent comic story generation. The platform enables users to create complete 5-panel comic stories from a simple text prompt, including AI-generated illustrations, narration captions, character dialogue, and professional PDF export.

Technical Architecture

ComicCraft uses a modular three-tier architecture:

Presentation Layer: Responsive dark glassmorphism UI built with HTML5, CSS3, and Vanilla JavaScript using Jinja2 templates for server-side rendering.

Application Layer: FastAPI backend orchestrating the AI pipeline - Gemini Flash generates outlines, Gemini Pro creates narratives, Stable Diffusion produces illustrations, and the layout builder assembles everything.

Data Layer: File-based storage (static/panels/ for images, static/exports/ for PDFs) with .env for secure API key management.

Pre-requisites

FastAPI Framework Knowledge: [FastAPI Documentation](#)

Google Gemini API Familiarity: [Google AI Studio / Gemini API Docs](#)

Hugging Face Diffusers: [Diffusers Documentation](#)

HTML, CSS, JavaScript: [W3Schools Tutorials](#)

Python Programming: [Python Documentation](#)

Version Control with Git: [Git Documentation](#)

FPDF2 Library: [PyPI - fpdf2](#)

Project Workflow

Milestone 1: Model Selection & Architecture Setup

Generate Gemini API key and configure .env

Research and select Gemini models (Flash for outlines, Pro for stories)

Define modular architecture and component interactions

Set up FastAPI project structure and dependencies

Milestone 2: Core AI Functionalities

Develop Gemini Flash outline generation (structured 5-panel JSON)

Develop Gemini Pro story/narration expansion

Integrate Hugging Face Diffusers for image generation

Milestone 3: Application Logic

Write FastAPI route handlers in routes.py

Implement pipeline orchestrator in pipeline.py

Add mock fallback mode for API-free testing

Milestone 4: Frontend Development

Design responsive input form (index.html)

Build interactive comic preview page (comic_preview.html)

Implement download/export flow

Milestone 5: PDF Export & Testing

Implement FPDF2 multi-page export with Unicode fonts

Write unit tests for all core modules

Test full pipeline with sample prompts

Milestone 6: Deployment & Documentation

Prepare local deployment with Uvicorn

Write comprehensive README

Create demo video and final submission